

MARKUS PETTERS

COMBUSTION EMISSIONS

Combustion

Combustion is high-temperature reaction that typically involves the oxidation of fuel. The reaction is exothermic, meaning the reaction releases energy. Many common fuels are carbon-based, including coal, gasoline, natural gas or wood. Complete combustion converts the fuel to carbon dioxide (CO_2) and water (H_2O). Incomplete combustion results in a number of other byproducts. One of the main products of incomplete combustion is carbon monoxide (CO). Other common products are nitrogen oxides (NO_x), formaldehyde (HCHO), and soot (light absorbing particulate matter). Fuel constituents other than carbon may also release pollutants. For example, fuel-bound sulfur may be oxidized during combustion to release sulfur dioxide (SO_2). More generally, combustion emissions depend on the fuel chemistry, the structure and arrangement of fuel (size, shape, and packing), the fuel condition (moisture content), and the combustion conditions (oxygen content, flaming vs. smoldering vs. glowing).

Pollutants produced by incomplete combustion are dispersed into the environment as a smoke plume. The plume is most concentrated near the source and diluted downwind. To account for dilution, emissions are commonly quantified as emission factors normalized by the amount of fuel burnt. This normalization is referred to as the carbon mass balance method.



Figure 1: Car exhaust plume.

Carbon Mass Balance Method

Emission ratios (ERs) quantify the abundance of pollutants in fresh plumes. Emission factors (EFs) quantify mass emitted per fuel burned. EFs are determined by measuring the concentration of pollutants in the background air, and the elevated concentration of pollutants inside the plume. The concentration difference $\Delta C = C_{plume} - C_{background}$ defines the excess concentration. The emission factor can be calculated from the excess concentration using the carbon mass balance method. The carbon mass balance assumes that the plume originates from the combustion of carbon-containing fuel. The vast majority of fuel is converted to either CO or CO_2 . Trace amounts are converted to other carbon containing species, e.g. HCHO or carbonaceous particulate matter. Other trace species such as NO_x or SO_2 may also be included.

The emission ratio of species X is denoted ER_x and calculated as the ratio of $\Delta C_x / \Delta \text{CO}_2$. The emission factor EF_x expresses the amount of pollutants produced per unit fuel combusted, and is often reported as $\text{g pollutant} / \text{kg fuel}$. It is calculated via

One of the main advantages of the fuel-based form is that measurements can be made without knowing vehicle speed or load. However, fuel-based EFs ($\text{g} / \text{kg fuel}$) cannot be directly compared to alternative standards that are also used, e.g. distance-based (g / mile) or energy-based (g / kWh) EFs .

$$EF_x = F_c \times 1000 \frac{g}{kg} \times \frac{MW_x}{12.011} \times \frac{ER_x}{C_T} \quad (1)$$

where F_c is the mass fraction of carbon in the fuel, MW_x is the molar mass of species X in $g/mole$, 12.011 is the molar mass of carbon in $g/mole$ and C_T is

$$C_T = \sum_{j=1}^n \frac{N_j \times \Delta C_j}{\Delta CO_2} \quad (2)$$

Here n is the number of carbon-containing species measured, N_j is the number of carbon atoms in species j , and ΔC_j is the excess mixing ratio of species j . ΔC_x and all ΔC_j must be in the same units.

Equations (1) and (2) require mixing ratios. However, particulate PM is reported as a mass concentration ($\mu g\ m^{-3}$) and must be converted first. In Equation (2) the product $N_j \times \Delta C_j$ converts from ppm of a molecule to $ppmC$, which is parts per million carbon or micro-moles of carbon atoms per mole of air. For example a mixing ratio of 0.2 ppm of C_3H_8 (propane) is 0.6 $ppmC$ since $N = 3$ carbon atoms per molecule. Mass concentration can be converted to $ppmC$ using the following method

$$C\ [ppmC] = f_{C,PM} \times \frac{RT}{M_C p} \times C\ [\mu g\ m^{-3}] \quad (3)$$

where $f_{C,PM}$ is the fraction of PM mass that is actually carbon atoms, R is the ideal gas constant, M_C is the molar mass of carbon, p is the pressure, and T is the temperature. Calculated at $T = 298.15K$ and $p = 1013.25\ hPa$:

$$C\ [ppmC] = f_{C,PM} \left[\frac{\mu gC\ m^{-3}}{\mu g\ m^{-3}} \right] \times C\ [\mu g\ m^{-3}] \times 0.00204 \left[\frac{ppmC}{\mu gC\ m^{-3}} \right] \quad (4)$$

For particulate matter the emission factor is then calculated as

$$EF_{PM} \left[\frac{g\ PM}{kg\ Fuel} \right] = F_c \left[\frac{g\ C}{g\ Fuel} \right] \times 1000 \left[\frac{g}{kg} \right] \times \frac{\Delta PM\ [\mu g\ m^{-3}]}{\sum N_j \Delta C_j\ [ppmC]} \times 0.00204 \left[\frac{ppmC}{\mu gC\ m^{-3}} \right] \quad (5)$$

Determination of the factor $f_{C,PM}$ requires PM composition measurements. The value is 0.8–0.85 for gasoline exhaust, 0.90–0.95 for diesel particulate matter, and 0.55–0.64 for wood smoke.

Worked Example

Emissions are measured by bisecting tailpipe plumes. Figure 2 shows a time series of NO , and CO_2 behind a gasoline vehicle. The mass

fraction of carbon in the fuel is $F_c = 0.85$. The dashed horizontal line is the baseline concentration. In the plume, the NO concentration is increased from 20 *ppb* to 420 *ppb*. The change in concentration $\Delta NO = 400$ *ppb*. At the same time, CO_2 increased from 427 *ppm* to 470 *ppm*. The change is $\Delta CO_2 = CO_{2,plume} - CO_{2,baseline} = 470 - 427$ *ppm* = 43 *ppm*. Now consider the table below, which includes similar data for other pollutants:

Quantity	Value	N_j
ΔCO_2	43 <i>ppm</i>	1
ΔNO	0.4 <i>ppm</i>	0
ΔCO	0.5 <i>ppm</i>	1
$\Delta PM_{2.5}$	1.1 $\mu g\ m^{-3}$	-
Particulate carbon	0.0018 <i>ppmC</i>	1

Note that NO contains no carbon, so $N_j = 0$. In the calculations all units need to be consistent and therefore 400 *ppb* is converted to 0.4 *ppm*. The conversion from $\Delta PM_{2.5}$ to particulate carbon calculated as $0.8 \frac{\mu g\ C\ m^{-3}}{\mu g\ m^{-3}} \times 1.1\ \mu g\ m^{-3} \times 0.00204\ \frac{ppmC}{\mu g\ C\ m^{-3}} = 0.0018\ ppmC$. The emission factors are calculated as follows.

1. Total excess carbon and C_T

$$\Sigma N_j \Delta C_j = 1 \times 43 + 1 \times 0.5 + 1 \times 0.0018 = 43.5018\ ppmC$$

$$C_T = 43.5018/43 = 1.0117$$

Note the relatively small contribution of ΔCO and particulate carbon to C_T . Neglecting everything but CO_2 gives $C_T = 1$ instead of 1.0117. Not accounting for the minor species would result in a 1.2% bias in the emission factors below.

2. CO Emission factor

$$ER_{CO} = 0.5/43 = 0.0116$$

$$EF_{CO} = 0.85 \times 1000 \times (28.01/12.011) \times (0.0116/1.0117) = 22.73\ g\ CO\ kg^{-1}$$

3. NO Emission factor

$$ER_{NO} = 0.4/43 = 0.0093$$

$$EF_{NO} = 0.85 \times 1000 \times (30.01/12.011) \times (0.0093/1.0117) = 19.52\ g\ NO\ kg^{-1}$$

4. PM Emission factor

$$EF_{PM} = 0.85 \times 1000 \times 1.1/43.5018 \times 0.00204 = 0.044\ g\ PM\ kg^{-1}$$

5. CO_2 Emission factor

$$EF_{CO_2} = 0.85 \times 1000 \times (44.01/12.011) \times (1/1.0117) = 3.078\ kg\ CO_2\ per\ kg\ fuel$$

Taking ΔC as (peak – baseline) for each species independently is the most noise-sensitive method and is biased whenever two sensors have different response times. Figure 2 illustrates the problem: the

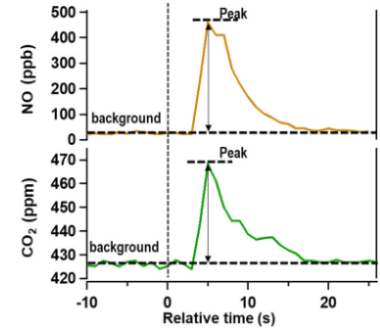


Figure 2: Time series of pollutant signals. The vertical dashed line indicates the time at which a vehicle passes the sensor platform. Source: Shen et al. (2022, <https://doi.org/10.1016/j.scitotenv.2021.151609>)

Note that C_T is built only from **measured** carbon species, so any unmeasured carbon (volatile organic compounds, CH_4) makes the sum $\Sigma N_j \Delta C_j$ too small and biases all emission factors high. For gasoline exhaust this is error is small but for wood smoke or smoldering combustion the unmeasured fraction can be several percent.

NO and CO_2 peaks have different shapes and decay rates due to instrument response time. One approach to overcome this limitation is to regress ΔX against ΔCO_2 over the whole plume transect and take the slope as ER . Equivalently, one can integrate both signals over the plume.

Important Gases and Pollutants

Carbon Dioxide

Carbon dioxide (CO_2) is a colorless and odorless gas. It is one of the most important atmospheric constituents that has many sources, sinks, and environmental effects. Sources of CO_2 include respiration, outgassing from oceans and volcanoes, biomass burning, and fossil fuel combustion. Sinks of CO_2 include photosynthesis, uptake by oceans, soils, and chemical weathering of rocks. Carbon dioxide is a greenhouse gas that traps heat and CO_2 is correlated with global mean temperature. In the indoor environment the buildup of CO_2 can be used as a tracer for air exchange rates between the enclosure and the outdoor environment.

Global CO_2 concentrations have increased from 280 ppm at the start of the industrial revolution in 1750 to 430 ppm in 2026. Photosynthesis in the northern hemisphere summer and respiration in the northern hemisphere winter are responsible for a clear annual cycle in CO_2 concentration. At high concentration, CO_2 is harmful to human health. The 8 hr personal exposure limit is set to 5,000 ppm . Some studies report at concentrations as low as 1,000 ppm , but this threshold is not universally accepted. Cognitive impairment and drowsiness will occur at 10,000 – 20,000 ppm . At > 20,000 ppm symptoms of headache, sleepiness, loss of attention, and nausea appear. Poor ventilation can cause the quick buildup of CO_2 concentration to dangerous levels. For example, a sedan with four passengers and the air handler set to recirculation mode will result in cabin CO_2 of 2,500 – 4,000 + ppm within an hour, caused by the respiration and exhalation CO_2 . During combustion, carbon based fuels are converted mostly to CO_2 . Gasoline and diesel fuels produce 3.05 – 3.2 kg for CO_2 per kg of fuel. Therefore, the excess of CO_2 in exhaust plumes can be used as a tracer to derive fuel-based emission factors for other pollutants.

The current status of CO_2 as a pollutant remains unclear. Pollution can be defined as anthropogenically emitted substances that build up in concentration large enough to cause direct or indirect damage to humans or ecosystems. In 2007 the U.S. supreme court held that greenhouse gases fall within the Clean Air Act's definition of "air



Figure 3: Skeletal formula of CO_2 .

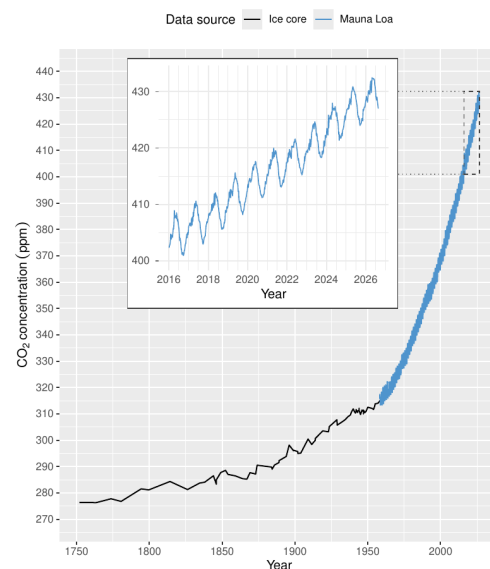


Figure 4: Time series of atmospheric CO_2 concentration from 1750 to present day.

pollutant" and that EPA must determine whether they endanger public health or welfare. In 2009 the EPA issued an endangerment finding that CO₂ harms public health and welfare because it contributes to climate change. The endangerment finding formed the legal basis to regulate emissions from fossil fuel use, including the setting of fuel efficiency standards. In 2025 repeal of the endangerment finding was proposed. EPA finalized the rescission of the 2009 endangerment finding on February 12, 2026, and also finalized repeal of greenhouse gas emission standards for on-highway vehicles and engines. The final rule was published in the Federal Register on February 18, 2026, took effect April 20, 2026. As of August 2026, litigation challenging it is expected to reach the Supreme Court. Globally, some regions have imposed various carbon taxes to reduce emissions. However, no binding global instrument caps emissions and total emissions remain de facto unregulated.

Carbon Monoxide

Carbon monoxide (CO) is a colorless and odorless gas. Its main anthropogenic source is from incomplete combustion. Other sources of CO include biomass burning, biological activity in vegetation and soils, photochemical breakdown of dissolved organic matter in oceans, and photochemical production from atmospheric hydrocarbons. Atmospheric sinks include atmospheric reaction of CO with the hydroxyl radical (90%) and uptake of CO to soils (10%). The oxidation reaction is slow, resulting in an atmospheric lifetime of CO from 2 weeks in the tropics to several months at high latitudes in winter. Its long atmospheric lifetime, combined with predominantly combustion sources makes CO a useful tracer to evaluate the transformation of polluted air as it is transported from the source of large distances and long time scales.

Carbon monoxide has strong negative effects on human health. The molecule binds 200 – 250× that of O₂, forming carboxyhemoglobin and reducing oxygen transport to the brain, causing chemical asphyxiation via impaired O₂ delivery. One hour exposure to 300 ppm causes headache, 800 ppm causes dizziness, vomiting, and convulsions, and higher concentrations are fatal. Malfunctioning indoor combustion sources, including gas stoves, gas heaters, or wood stoves result in numerous fatal CO poisonings annually. Certified carbon monoxide alarms (UL 2034) are programmed to trigger if CO concentration exceeds 400 ppm for 4 – 15 min and 70 ppm for 60 – 240 min. In the 1970s and 1980s, atmospheric CO concentration sometimes exceeded 100 ppm on freeways and in tunnels. Inside vehicle cabins, CO concentration could reached ranges between 10 and

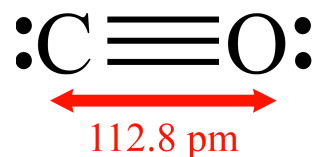


Figure 5: Skeletal formula of CO.

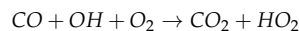


Figure 6: Carbon monoxide oxidation by the hydroxyl radical.



Figure 7: Example indoor CO detector.

60 *ppm*. Due to effective catalytic converter technology, U.S. urban CO has fallen to values that are typically well under 1 *ppm*. In the remote free troposphere, concentrations between 0.05 and 0.2 *ppm* are to be expected. Carbon monoxide is one of the six criteria pollutants regulated by the National Ambient Air Quality Standard (NAAQS). The primary 1 hr standard is 35 *ppm*, the 8 hr standard 9 *ppm*, not to be exceeded more than once per year.

Since CO is produced by incomplete combustion, CO can be used to characterize combustion efficiency. The modified combustion efficiency (MCE) is defined as the ratio of excess CO₂ and the sum of excess CO and CO₂ concentration. Values > 0.99 indicate near complete oxidation, which occurs for efficient gas burners. MCE < 0.75 corresponds to heavily oxygen-starved smoldering. Most emission factors vary strongly with MCE, with lower MCE corresponding to larger emission factors.

Nitrogen Dioxide

Nitrogen dioxide (NO₂) is a highly reactive air pollutant. NO₂ is one component of NO_x = NO + NO₂. The NO : NO₂ ratio varies with plume age and photochemistry. NO₂ is a reddish-brown light absorbing gas that is both an irritant and toxic. The lethal concentration for one hour of exposure for humans is ≈ 170 *ppm*. Environmental concentrations are much lower. The largest concentrations are measured curbside at 20 – 50 *ppb*. Urban background concentrations are 10 – 25 *ppb*, rural concentrations are 2 – 8 *ppb*, remote continental/free tropospheric concentrations 0.5 – 2 *ppb*, and remote marine concentrations are 0.02 – 0.5 *ppb*. NO₂ is one of the six EPA regulated criteria pollutants. The primary 1 hr standard defined in the National Ambient Air Quality Standard (NAAQS) is 100 *ppb*. Sources of NO_x include fossil fuel combustion, biomass burning, lightning, and microbial activity following agricultural fertilization. NO_x plays an important role in ozone and PM_{2.5} formation. Health effects of atmospheric NO_x include the irritation of airways in the human respiratory system and are linked to respiratory symptoms coughing, wheezing or difficulty breathing, may contribute to the development of asthma and may increase susceptibility to respiratory infections. The radical terminating reaction between NO₂ and the hydroxyl radical (OH) results in either nitric acid (HNO₃) or peroxyxynitrous acid (HOONO) formation. Nitric acid contributes to acid rain and ecosystem eutrophication.

Combustion produces NO_x by three mechanisms, thermal NO_x, fuel NO_x, and prompt NO_x. Thermal NO_x is formed at high temperature and converts atmospheric molecular nitrogen (N₂) to nitric

$$MCE = \frac{\Delta CO_2}{\Delta CO_2 + \Delta CO}$$

Figure 8: Definition of modified combustion efficiency.

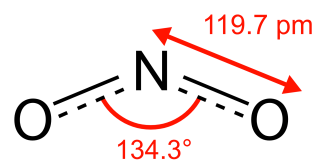


Figure 9: Skeletal formula of NO₂.



Figure 10: Plume of NO₂ released from a mining blast near Moranbah in 2020.

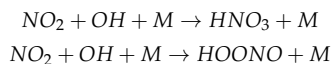


Figure 11: Radical terminating reactions leading to the removal of OH and NO₂ from the atmosphere.

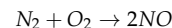
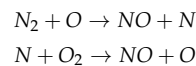


Figure 12: Zeldovich mechanism of thermal NO_x production.

oxide (NO) through the Zeldovich reaction mechanism. The rate of thermal NO_x formation is significant at temperatures exceeding 1800K, and NO_x formation increases with increasing temperature. Fuel NO_x arises by converting fuel bound nitrogen to NO . Heating of the fuel causes thermal decomposition of nitrogen containing compounds. In the reaction zone, nitrogen containing free radicals are formed (hydrogen cyanide radical HCN , cyano radical CN , nitrene radical NH), which are either oxidized to NO or reduced to N_2 .

NO_x formation is observed even when the thermal NO_x and the fuel NO_x mechanisms are eliminated. This pathway is called prompt NO_x and caused by the reaction of atmospheric nitrogen with combustion radicals. Prompt NO_x is most prevalent at low-temperature, fuel-rich conditions encountered in surface burners and gas turbines.

It is clear from the reactions that NO_x is formed as NO during combustion. A fraction of the produced NO is oxidized to NO_2 downstream of the combustor. At the point of emission 90-95% of the NO_x is in the form of NO for spark-ignition gasoline. Diesel engines with oxidation catalysts emit lower NO fractions. As NO enters the atmosphere it is converted to NO_2 by ozone within a few minutes. This finding has several important consequences. First, near emission sources ozone is often depleted, sometimes resulting in an apparent negative emission factor for ozone. Specifically this is caused by ΔO_3 being negative in fresh plumes. Second, NO emitted from the morning rush hour often depletes ozone and this delays the onset of a visible increase in ozone concentration toward later in the day. Finally, using an NO_2 sensor for emission factor measurements will strongly underestimate NO_x emissions, and emission factors will vary with the smoke age after emission.

Sulfur Dioxide

Sulfur dioxide (SO_2) is a colorless gas that plays an important role in air pollution. It has a strong odor at concentrations $> 0.5 \text{ ppm}$. The personal exposure limit set by the U.S. occupational safety and health administration (OSHA) is 5 ppm . SO_2 is water soluble and absorbed in the mucus membrane. Short term exposure stresses the respiratory system and is especially harmful for children and people with asthma. Long-term exposure has been associated with impaired lung function. Environmental concentrations are typically much lower than the OSHA limit, ranging from $1 - 30 \text{ ppb}$ in polluted air and $0.01 - 1 \text{ ppb}$ in the background troposphere. Sulfur dioxide is one of the six EPA regulated criteria pollutants. The primary 1 hr standard defined in the National Ambient Air Quality Standard (NAAQS) is 75 ppb . Sources of SO_2 include the oxidation of dimethyl

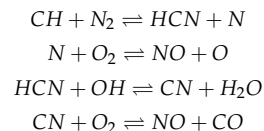


Figure 13: Prompt NO_x mechanism.

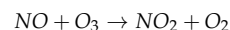


Figure 14: The main NO titration reaction describes the chemical destruction of nitric oxide by ozone or vice versa.

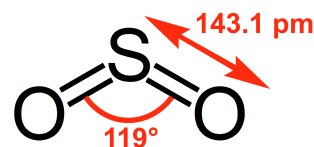


Figure 15: Skeletal formula of SO_2 .

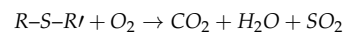


Figure 16: Schematic net reaction of fuel-bound organosulfur compounds to form sulfur dioxide. Note that the conversion is quantitative enough such that the SO_2 emission factor from combustion can be determined from fuel composition alone and technically needs no plume measurement.

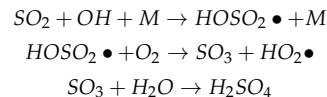


Figure 17: Photooxidation mechanism of sulfur dioxide by hydroxyl radicals to form sulfuric acid. The first step involves gas-phase oxidation of sulfur dioxide by hydroxyl radicals to form the intermediate hydroxysulfonyl radical, which is further oxidized by molecular oxygen to form sulfur trioxide, which then undergoes hydration to produce sulfuric acid.

sulfide emitted by oceanic phytoplankton, volcanic emissions, fossil fuel combustion, mineral ore processing, and chemical manufacturing. The main anthropogenic source is the oxidation of fuel-bound sulfur during combustion. The EPA Tier 2 and Tier 3 fuel standards program strongly reduced the fuel sulfur content, thus almost eliminating tailpipe SO_2 emissions over the last 30 years. Sulfur dioxide released into the air is converted to sulfuric acid through photochemical oxidation in the clear atmosphere and through aqueous phase oxidation in cloud water. Sulfuric acid contributes to particulate matter ($\text{PM}_{2.5}$) and acid rain which harms ecosystems.

Particulate Matter

Suspended particulate matter consists of solid or liquid particles with diameters between 1.5 nm and $100 \text{ }\mu\text{m}$. Particulate matter is regulated through the National Ambient Air Quality Standard (NAAQS) in two categories: PM_{10} , particulate matter less than $10 \text{ }\mu\text{m}$ in aerodynamic diameter and $\text{PM}_{2.5}$, particulate matter less than $2.5 \text{ }\mu\text{m}$ in aerodynamic diameter. The current primary 24 hr standard for $\text{PM}_{2.5}$ and PM_{10} is $35 \text{ }\mu\text{g m}^{-3}$ and $150 \text{ }\mu\text{g m}^{-3}$, respectively. The primary standard for the annual mean, averaged over 3 years is $9 \text{ }\mu\text{g m}^{-3}$.

Atmospheric particulate matter originates from a mixture of primary and secondary sources. Primary sources include combustion, dust from road, brake, and tire wear, wind-blown mineral dust, sea-spray, and biological particle emissions. Secondary sources include the photochemical production of low volatility chemicals that condense on preexisting particles. Precursors for secondary particulate matter are SO_2 , NO_x , and a large number of organic compounds. Example organic precursors of anthropogenic origin are evaporated solvents, evaporated gasoline fuel, and gas-phase emissions from fossil fuel combustion or biomass burning. Organic precursors of natural origin are evaporated compounds from living organisms, including dimethyl sulfide from phytoplankton or terpenes from deciduous and coniferous trees.

Particulate matter has a wide range of health and environmental effects. Exposure to elevated $\text{PM}_{2.5}$ and PM_{10} has been linked to respiratory symptoms, decreased lung function, and cardiovascular disease. Epidemiological studies have demonstrated increased mortality with increased PM exposure. Extreme exposure from inhalation of dense smoke, tobacco products, coal dust, silica dust, or diesel particulate matter leads to acute and chronic disease, e.g. coal miner lung disease and lung cancer. Inhalation of fibrous solid particles, e.g. those released by Asbestos, has particularly nasty health

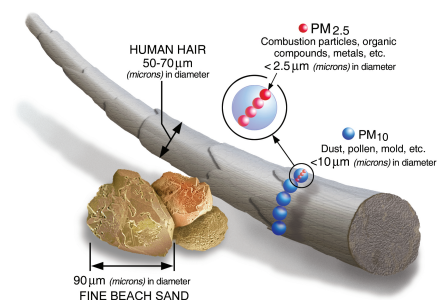


Figure 18: Illustration of the spatial scale of $\text{PM}_{2.5}$ and PM_{10} .



Figure 19: Direct emission of Diesel particulate matter.

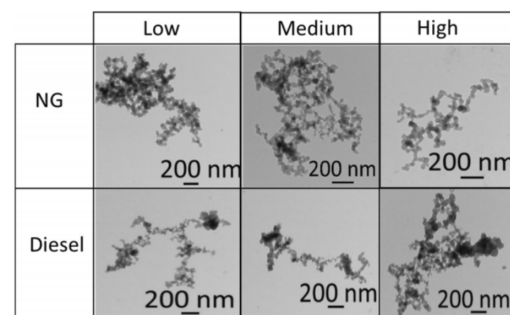


Figure 20: Images of typical soot aggregates generated by an engine operated with natural gas (top row) and on diesel fuel (bottom row). Source: Baldelli et al. (2020, <https://doi.org/10.4209/aaqr.2020.01.0006>).

effects. Fibrous solid particles have large length-to-width ratios (e.g. $10\ \mu\text{m} \times 200\ \text{nm}$ or 50:1) and are light enough to count as $\text{PM}_{2.5}$ or PM_{10} . They are aerodynamically transported to the alveoli where they deposit. Due to the elongated shape, macrophages cannot remove the intruders. Furthermore, because the particles are solid and water insoluble, they become biopersistent, resulting in inflammation and eventually disease. Current scientific evidence suggests that there is no safe level of exposure where PM stops causing harm.

Environmental effects of PM includes visibility impairment and damage to ecosystems. Visibility impairment is evident in regions of persistent photochemical smog, where secondary aerosol production leads to elevated PM concentration. Visibility impairment is a nuisance because it spoils the natural scenery. It also interferes with communication, aviation safety, and military operations. Ecosystem impacts of PM includes reduced crop yields, soil acidification, as well as lake and coastal water eutrophication. Light-absorbing PM reduces planetary albedo and warms the atmosphere and when deposited on snow or ice accelerated melting. Non-light absorbing particles increase in planetary albedo by directly reflecting solar radiation and by modifying cloud microphysics, and may alter rainfall patterns through indirect effects on clouds.

Gas Measurement

Research Grade Instruments

Real-time gas measurement systems can be broadly divided into two subsets: research grade instruments and industrial grade sensors. Research grade instruments for CO_2 , O_3 , SO_2 , NO_x and organic gases typically use spectroscopic or mass spectrometry quantification. Spectroscopic methods pass light of a specific wavelength through the gas sample (an absorption/optical cell) and is detected by a photodetector. Attenuation due to gas absorption at that wavelength is used to quantify the amount of gas. Spectroscopic methods often require long optical path lengths, reference cells to subtract the air background, and careful calibration of the signal. Mass spectrometers measure concentration by first ionizing the gas and then measure the abundance of the ion. Mass spectrometers require high vacuum and expensive detectors. The cost of research grade instruments ranges from \$5k to \$15k for some of the more easily measured gases to > \$100k for high quality measurement of organic gases. In addition to high cost, research grade instruments are often bulky and require specialized personnel to operate the equipment. With research grade equipment, gas concentration can be measured accurately, often at



Figure 21: Photochemical smog and visibility reduction in the San Bernardino valley viewed from Riverside.



Figure 22: Picarro research grade cavity ring-down spectroscopy CO_2 analyzer 380–2000 ppm at 0.2 ppm precision.

sub-ppb levels present in the remote environment.

Sensors on a Chip

Sensors on a chip are available for a number of important gases. Some of these sensors use spectroscopic methods (e.g. CO_2) but most exploit electrochemical reactions. Compared to research grade instruments, chip-based sensors have higher (worse) detection limits, lower (worse) measurement resolution, and slower (worse) time-response characteristics. However, chip-based sensors have a number of advantages that makes their use attractive, even for research purposes. First, these sensors are much lower in cost, ranging from $< \$10$ to a few hundred dollars per sensor. Second, chip-based sensors have a small footprint and low power requirements, allowing them to be deployed on battery power for extended periods of time. Low cost, low power, and a small physical footprint recommend these sensors for industrial monitoring, mobile applications, and deployment as wearable sensors. Battery powered personal CO_2 monitors that became popular during the COVID era are an excellent example. Chip-based sensors find widespread application in industry, homes, and environmental sensing. Carbon monoxide alarms found in most homes are another example of chip-based applications.

Electrochemical sensors work on the basis of oxidation-reduction reactions. The target gas enters the housing through a diffusion controlled inlet. It then reaches a sensing electrode where the target gas is oxidized or reduced. This induces a current that is proportional to the gas concentration. The current needs to be amplified and digitized by an electronic device that is then interfaced with a computer that reads the signal. Different target gases use different electrode materials. Typical catalysts use precious metals (e.g. gold) deposited on a substrate. Electrode design is based on target gas selectivity and sensitivity. Selectivity is a major concern with electrochemical sensors, with catalysts often responding to multiple gases.

Relative humidity may alter the reaction. At low relative humidity sensors may dry out. At high relative humidity electrodes may absorb extra water, altering reaction rates. Some sensors require a stabilized biased voltage and require up to 24 hr uptime before reaching their baseline current. The baseline current must be subtracted to zero the sensor. Unfortunately, the baseline current may also drift with sensor temperature and age. Because of the cross-sensitivity of analytes and baseline drift, electrochemical sensors can require complex algorithmic corrections to ensure high measurement accuracy. Nevertheless, electrochemical sensor current is highly linear with concentration and can, if properly calibrated, provide high accuracy

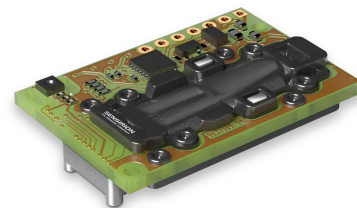


Figure 23: Sensirion SCD30 CO_2 sensor on a chip. The sensor uses non-dispersive infrared (NDIR) spectroscopy. Broadband IR is passed through the air and the intensity ratio of $3.9 \mu\text{m}$ and $4.26 \mu\text{m}$ transmitted light is proportional to concentration. The accuracy is $\pm(30 \text{ ppm} + 3\% \text{ of reading})$ over 400–10,000 ppm, which is sufficient for indoor air quality but would be insufficient to capture the shorter term trends at Mauna Loa.

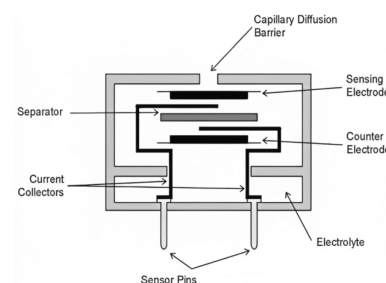


Figure 24: Electrochemical sensor structure diagram. The target gas diffuses to the working electrode where it is oxidized or reduced, resulting in current between the working electrode and counter electrode. The current is passed to the sensor pin, which connects to a chip where the current is amplified and made readable to a computer.



Figure 25: Example electrochemical cell and chip.

data in the ppb and ppm concentration range.

Particulate Matter Measurement

The reference method for measuring $PM_{2.5}$ and PM_{10} is conceptually quite simple. Air is sampled through an inlet that passes particles with diameter $< 2.5 \mu m$ and $< 10 \mu m$ in aerodynamic diameter. Sample flow is directed through a fibrous filter where particles deposit. Mass of the filter is measured before and after sample collection using and environmentally stabilized microbalance. Gravimetric filter analysis is highly accurate but also slow and labor intensive. Sample must be accumulated over an extended period of time, sometimes spanning several days. Filters must be handled by technicians and transported from the field to a laboratory for weighing. Faster methods are desirable to capture time evolution (e.g. the diurnal cycle) and episodic emissions. A beta attenuation monitor collects particles on a rotating filter and measures the attenuation of β -radiation emitted from a radioactive ^{14}C source. It is one of the EPA recognized equivalence methods and widely used to operationally monitor $PM_{2.5}$ and PM_{10} within an air shed.

Optical particle size spectrometers offer a relatively inexpensive method to measure the particle size distribution between $\approx 0.1 \mu m$ and $30 \mu m$. Integration over the size distribution gives estimates for $PM_{2.5}$ and PM_{10} . The relationship between optically-derived and filter-based PM measurements is approximate for two reasons. Optical particle counters are insensitive to ultrafine particles with diameter $< 0.1 \mu m$. Their contribution to the mass concentration will be missed. However, most of the $PM_{2.5}$ mass is contributed by particles larger than $0.1 \mu m$ in diameter, so this limitation is not too serious. More important is the distinction between optical and aerodynamic diameter. Aerodynamic diameter refers to the behavior of particles in accelerating flow fields. Aerodynamic diameter is sensitive to particle density. Optical diameter refers to the amount of light scattered when a particle transits a laser beam. The relative intensity of the light scattered depends on the particle shape and chemical composition. Light absorbing materials, like those found in diesel particulate matter, are particularly problematic for optical particle size spectrometers. Furthermore, light scattering of submicron particles is complex, resulting in non-linear scattering vs. diameter response that fundamentally limits the accuracy to which an individual particle can be sized using optical methods.

Advantages of optical spectrometers are their relatively small footprint, the ability to measure the size of individual particles, to measure the number concentration of particles in air, to measure par-



Figure 26: MetOne Environmental Beta Attenuation Mass (E-BAM) Monitor. The inlet removes particle with diameter $> 2.5 \mu m$. Particles are deposited on a rotating filter tape. Radiation is measured for the filter blank and sample filter. The difference in radiation is proportional to mass concentration.



Figure 27: GRIMM portable optical particle size spectrometer.

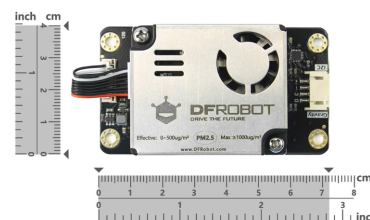


Figure 28: DFROBOT low cost air quality sensor.

ticulate matter at very low concentration (e.g. for clean room applications), and to measure at faster than 1 Hz time resolution. Optical particle size spectrometers measure short pulses of weak intensity and microsecond duration, which requires non-trivial signal amplification and electronics. Therefore the cost of optical spectrometers remains substantial, typically costing > \$15k per unit.

Over the last two decades, low cost air quality sensors that purportedly measure PM_1 , $PM_{2.5}$ and PM_{10} have flooded the market. These sensors use operating principles similar to optical particle size spectrometers. Unlike optical spectrometers, which measure the air sample particle-by-particle, low cost air quality sensors measure the light scattering of the particle ensemble with the optical cell to infer particle mass concentration. The mapping of the measured signal to the mass concentration in the PM_1 , $PM_{2.5}$ and PM_{10} size ranges is generally not disclosed by the manufacturer and requires making assumptions about the size distribution and chemical composition. Like gas sensors on a chip, low cost air quality sensors can qualitatively and semi-quantitatively detect particulate matter, albeit at reduced sensitivity and accuracy compared to research grade methods. Reliably integrating data from community owned and spatially distributed air quality sensor networks is one of the current challenges in environmental engineering.

Measurement Uncertainties

Measurement uncertainties impose significant constraints on the experimental design of emission factor measurements with chip-based sensors. The example shown in Figure 2 was obtained using research grade instruments. Resolving $\Delta CO_2 = 43 \text{ ppm}$ is challenging with a chip based CO_2 sensor that is only accurate $\pm 30 \text{ ppm}$. More concentrated plumes must be measured to reduce uncertainty, preferably $\Delta CO_2 > 400 \text{ ppm}$. Other sensors may also struggle to measure above their lower limits of detection. Within plumes, the detection efficiency of all sensors may differ from published performance characteristics. For example, the PM sensor maps an optical signal to mass concentration that is optimized for ambient particulate matter. The mapping may or may not hold for emission plume particulate matter, which most certainly will have a different size distribution and light absorbing properties compared to ambient PM . Cross-sensitivity of electrochemical sensors might falsely amplify or suppress the apparent measured EFs . Values obtained with non research grade equipment must be interpreted with caution when comparing to published values. Nevertheless, the EF measurements will provide relative bounds on the type and quantity of emissions from different sources, which

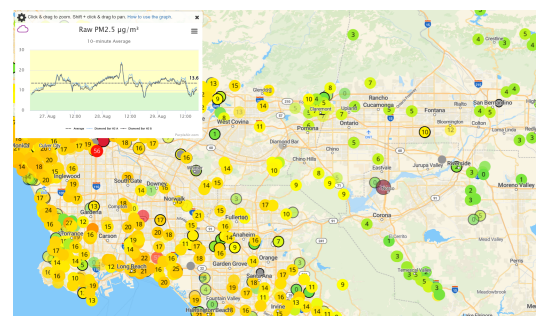


Figure 29: Example PurpleAir community sensor map (<https://map.purpleair.com/>). PurpleAir sells low cost air quality sensors that are networked and share data through the portal.

provides valuable insights into air quality processes.